Algebra II with Trigonometry Chapter 5.5

Olympiad Solutions

Saved tutoring response with MathJax rendering. Original PDF file: 4465374_alg-2trig-h-chp-5-5-note-docx-1.pdf

Solutions

---

1. In triangle ABC, angle A is twice angle B and side a is twice side b. What is the value of angle C?

Let angles be A, B, C (in degrees), and sides opposite them be a, b, c. We are told $A = 2B$, $a = 2b$.

By Law of Sines:

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

Given $a = 2b$, $A = 2B$, so:

$$\frac{2b}{\sin 2B} = \frac{b}{\sin B}$$

$$\Rightarrow \frac{2}{\sin 2B} = \frac{1}{\sin B}$$

$$\Rightarrow 2\sin B = \sin 2B$$

Recall $\sin 2B = 2\sin B \cos B$:

$$2\sin B = 2\sin B \cos B$$

$$\Rightarrow \sin B(1 - \cos B) = 0$$

Either $\sin B = 0$ (impossible for a triangle), or $\cos B = 1$ (then $B = 0°$), both invalid.

But let's check for $\cos B = 1$—not allowed. This means our earlier steps may need a re-examination.

Let's try a different view:

From the Law of Sines and given $a = 2b$, $A = 2B$:

$$\frac{a}{b} = \frac{\sin A}{\sin B} = \frac{\sin 2B}{\sin B} = \frac{2\sin B\cos B}{\sin B} = 2\cos B$$

But $a/b = 2$, so:

$$2\cos B = 2 \implies \cos B = 1 \implies B = 0^\circ$$

Again impossible.

Why? If you try in radians, identical conclusion. So, **no such triangle exists**.

However, the intent of the problem suggests there should be a solution; perhaps the constants are more subtle.

Suppose $A = 2B$ and $a = 2b$. Try to set up sum of angles:

$$A + B + C = 180^\circ \implies 2B + B + C = 180^\circ \implies C = 180^\circ - 3B$$

From Law of Sines:

$$\frac{a}{\sin A} = \frac{b}{\sin B} \implies \frac{2b}{\sin 2B} = \frac{b}{\sin B} \implies \frac{2}{\sin 2B} = \frac{1}{\sin B} \Rightarrow 2\sin B = \mathrm{si}$$

So either $\sin B = 0$ (impossible), or $\cos B = 1$ (impossible).

Thus, there is no triangle with these properties.

Answer: $\boxed{\text{No such triangle exists; the problem has no solution.}}$

2. In triangle PQR, PQ = 5, QR = 7, and angle Q = 60°. Without using a calculator, determine whether two distinct triangles can be constructed with these parameters. Justify mentally.

Let triangle PQR have sides $PQ = 5$, $QR = 7$, and angle $Q = 60^\circ$. Assume $PR = x$.

Given two sides and a non-included angle: this is the ambiguous case (SSA).

Let's "swing" side PQ from Q at 60°: depending on length, get 0, 1, or 2 triangles.

Let's call $a = QR = 7$, $b = PQ = 5$, and angle $Q = 60^\circ$ is between them.

But which sides are adjacent? Label Q between P and R, so

- PQ and QR meet at Q, angle at Q.

Use Law of Sines:

$$\frac{PQ}{\sin R} = \frac{QR}{\sin P}$$

But easier is to use the fact that the side opposite the given angle (Q) is not given.

To check for ambiguous case, assign the known quantities:

Let's try to use Law of Cosines to see if the third side is uniquely determined.

Let

- Q is at the known 60°,
- Sides $PQ = 5$, $QR = 7$, enclosed angle 60°.

So the triangle is determined completely by two sides and the included angle.

In such a case, **only one triangle can be constructed** (SAS). There is no ambiguity, since the angle is between the given sides.

Answer:

$\boxed{\text{Exactly one triangle can be constructed; there is no ambiguity since the giv}}$

---

3. Let triangle XYZ have sides $x = 8, y = 15$, and angle $X = 30^\circ$. Show mentally whether a triangle exists, and if so, whether more than one triangle is possible (the ambiguous case).

SSA case: sides $x = 8$, $y = 15$, angle $X = 30^\circ$ (angle opposite side $x = 8$).

Apply Law of Sines to check possible number of triangles:

$$\frac{x}{\sin X} = \frac{y}{\sin Y} \implies \frac{8}{\sin 30^\circ} = \frac{15}{\sin Y}$$

But $\sin 30^\circ = 1/2$:

$$\frac{8}{1/2} = 16 = \frac{15}{\sin Y} \implies \sin Y = \frac{15}{16} < 1$$

So:

- $\sin Y < 1$ and $0 < \sin Y < 1$ ⇒ **two possible values for angle Y**:
- $Y_1 = \arcsin(15/16)$
- $Y_2 = 180^\circ - Y_1$

But the sum of the two known angles $X + Y_2$ must be < 180°. Let's check:

- $Y_2 = 180^\circ - \arcsin(15/16)$
- $X = 30^\circ$
- If $30^\circ + Y_2 = 210^\circ - \arcsin(15/16) > 180^\circ$ always, so only Y_1 possible.

Thus, **only one triangle exists** (the ambiguous solution with obtuse angle is not possible).

Answer:

$\boxed{\text{Exactly one triangle exists (no ambiguity); } \sin Y = 15/16 < 1, \text{ but the "sw}}$

---

4. Triangle DEF has angles D, E, F such that $\sin D + \sin E = \sin F$. If sides d, e, f are opposite these angles respectively, what can you say

about the ratios $d : e : f$?

Law of Sines:

$$\frac{d}{\sin D} = \frac{e}{\sin E} = \frac{f}{\sin F} = 2R \implies d : e : f = \sin D : \sin E : \sin F$$

Given: $\sin D + \sin E = \sin F$. Thus:

$$d : e : f = \sin D : \sin E : (\sin D + \sin E)$$

Answer: $\boxed{d : e : f = \sin D : \sin E : (\sin D + \sin E)}$

---

5. In triangle LMN, $LM = 10$, $MN = 7$, and angle $N = 90^\circ$. Find $\sin L + \sin M$ using only properties of sines in a triangle.

Let's label sides:

- Opposite L, M, N are l, m, n respectively.
- So: angle at $N = 90^\circ$.
- LM is the side opposite $N \Rightarrow n = LM = 10$.
- The other known side: $MN = 7$, which is side l (opposite L)? By order of letters, yes.
- MN is side l, so $l = 7$.

Use Law of Sines:

$$\frac{n}{\sin N} = \frac{l}{\sin L}$$

$\sin N = 1$ since $N = 90^\circ$.

$$\frac{10}{1} = \frac{7}{\sin L} \implies \sin L = \frac{7}{10}$$

Similarly, for angle M, side m is opposite M, but not given directly.

But $\sin L + \sin M = ?$

But since $L + M = 90^\circ$, as $N = 90^\circ$.

Therefore:

$$\sin L + \sin M = \sin L + \sin(90^\circ - L) = \sin L + \cos L$$

We have $\sin L = 0.7$. Find $\cos L$:

$$\cos L = \sqrt{1 - \sin^2 L} = \sqrt{1 - (0.7)^2} = \sqrt{1 - 0.49} = \sqrt{0.51}$$

Thus,

$$\sin L + \sin M = 0.7 + \sqrt{0.51}$$

But the question says "using properties of sines", i.e., just express in terms of $\sin L$:

$$\boxed{\sin L + \sin M = \sin L + \cos L}$$

So, using the sides:

$$\sin L = \frac{7}{10}$$

$$\sin M = \cos L = \sqrt{1 - (\frac{7}{10})^2} = \sqrt{\frac{51}{100}} = \frac{\sqrt{51}}{10}$$

Therefore,

$$\sin L + \sin M = \frac{7}{10} + \frac{\sqrt{51}}{10} = \frac{7 + \sqrt{51}}{10}$$

Answer: $\boxed{\sin L + \sin M = \frac{7 + \sqrt{51}}{10}}$

---

6. Triangle ABC has sides $a = \sqrt{3}, b = 1$, and angle $C = 120^\circ$. Find, without explicit calculation, whether triangle ABC is isosceles, scalene, or impossible.

Let sides a, b, c be opposite angles A, B, C.

Given: $a = \sqrt{3}$, $b = 1$, and angle $C = 120^\circ$.

Let’s use Law of Cosines to see if c is possible, and to deduce side lengths.

Law of Cosines:

$$c^2 = a^2 + b^2 - 2ab\cos C$$

But to judge isosceles-ness, let's check if any two sides can be equal, or if any triangle exists.

First, since C is the largest angle, side c should be the largest side.

Compare $a = \sqrt{3} \approx 1.732$, $b = 1$:

Is it possible for side $c > a, b$? Let's compute:

$$c^2 = a^2 + b^2 - 2ab\cos 120^\circ$$

$\cos 120^\circ = -1/2$:

$$c^2 = (\sqrt{3})^2 + 1^2 - 2 \cdot \sqrt{3} \cdot 1 \cdot (-1/2) = 3 + 1 + \sqrt{3} = 4 + \sqrt{3} > 0$$

Therefore $c = \sqrt{4 + \sqrt{3}}$, which is larger than a and b.

Now, compare side lengths:

- $c > a > b$ (distinct).
- All sides are different ⇒ **scalene**.

Answer:

$\boxed{\text{The triangle is scalene, since all three side lengths are distinct and positive.}}$